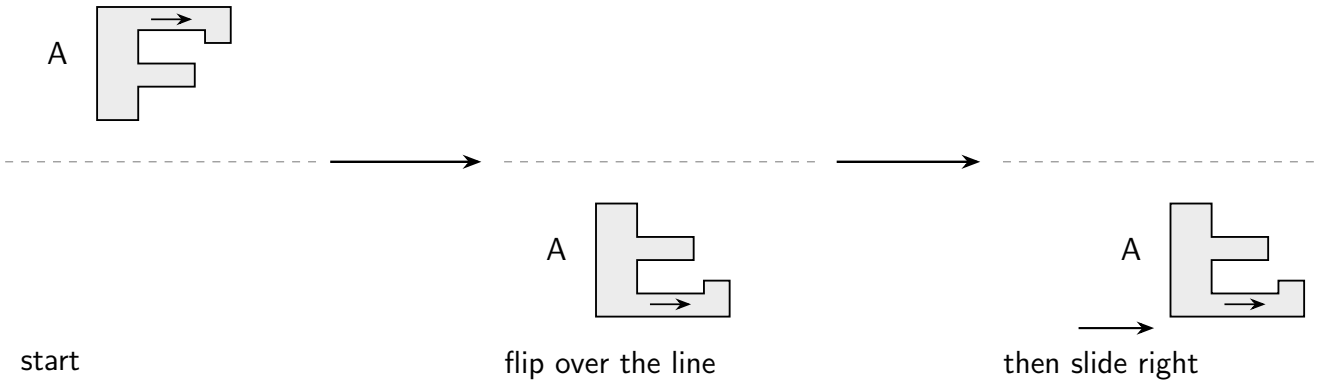
Use the loose footprints. Build each border in its box, or on a long strip if it needs more room. Each border repeats forever both ways. A match must fit every footprint. Only footprints count, not strip ends or guide lines. A slide must move the copy. Test with a tracing copy; leave the original in place.

Problem 1: Build two repeating borders with different shortest matching slides. Measure those slides; explain why a shorter slide cannot match each border.

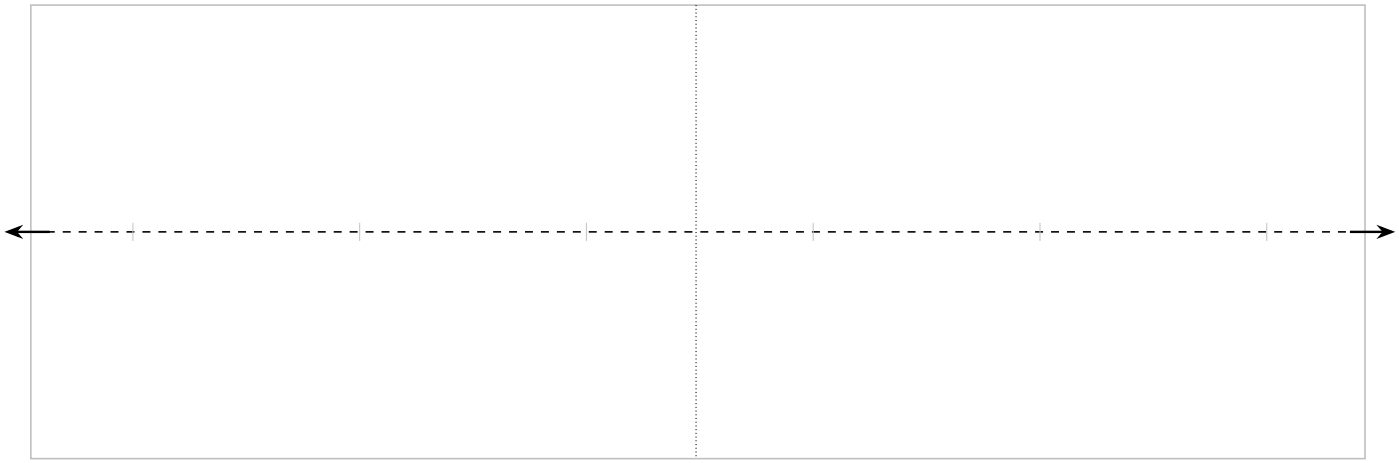


Problem 2: A glide is a flip over the middle line followed by a nonzero slide along it. Build a border that matches after a glide, but fails after the flip alone. Find its shortest matching slide and shortest matching glide.

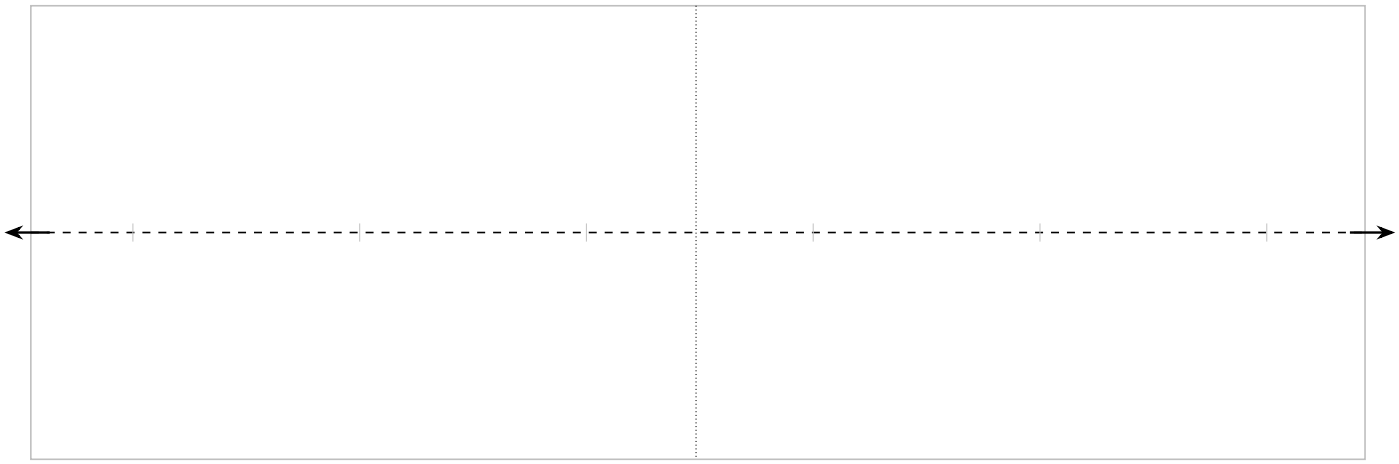
Extra workspace

Problem 3: A border matches after a 3 cm glide. Must a 6 cm slide also match? Must a 3 cm slide match? Explain why.

Problem 4: Make a repeating border that matches after a flip over each of the two crossing lines. Which other matching moves are forced by those two flips? Explain why.



Extra workspace



A half-turn turns the copy halfway around a chosen point on the table. You may choose any point.

Problem 5: Make a border that matches after a half-turn and fails after a flip over the middle line. Can you also make every flip across a line perpendicular to the border fail?

Extra workspace

Problem 6: For one border you made, describe every matching slide, glide, flip and turn. Explain why no other move matches.